

Enhancing ChEMBL: Integrating Drug and Clinical Candidate Data

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Overview

- Chemical biology resources (EMBL-EBI) contain a wealth of open-access preclinical and drug discovery bioactivity data
- Overview of the ChEMBL database and why drug data curation matters
 - ChEMBL contains drug discovery data
 - How is drug integration done?
 - The Clinical Trials Pipeline
 - Drug features in ChEMBL (indication, mechanism of action, drug properties, etc)
- Current Challenges

Open-access Chemical Biology Resources at EMBL-EBI



ChEMBL

*Bioactivities
and drug data*



SureChEMBL

*Structures,
targets and
diseases from
patents*



UniChem

*Linking
databases via
InChI*

website: <https://www.ebi.ac.uk/chembl/>

helpdesk: chembl-help@ebi.ac.uk



ChEBI

*Chemical
ontology, and
chemical
registration db for
biological dbs*

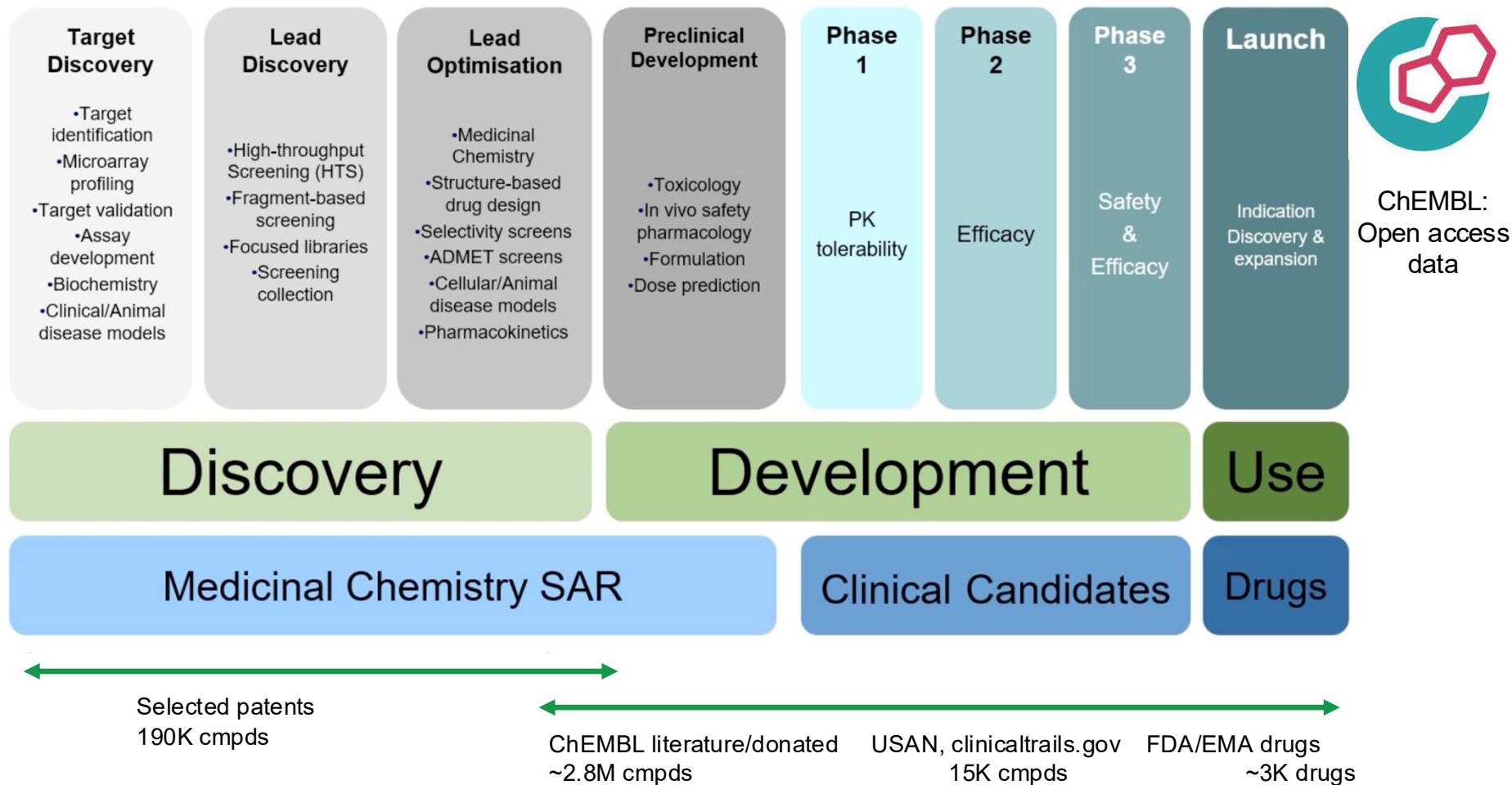


OPSIN

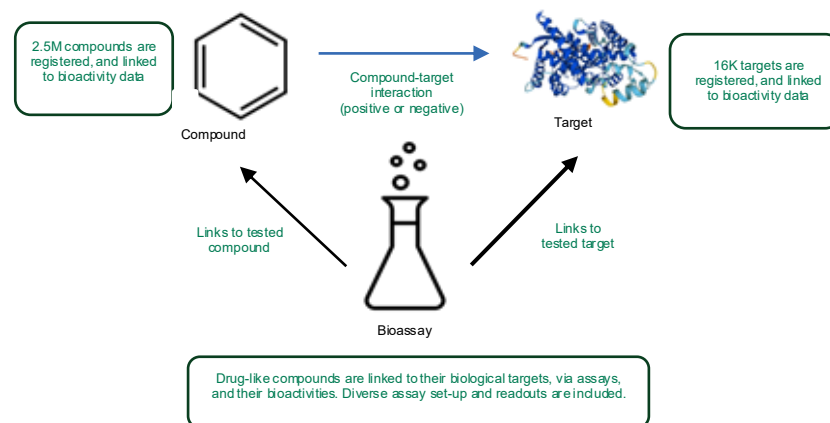
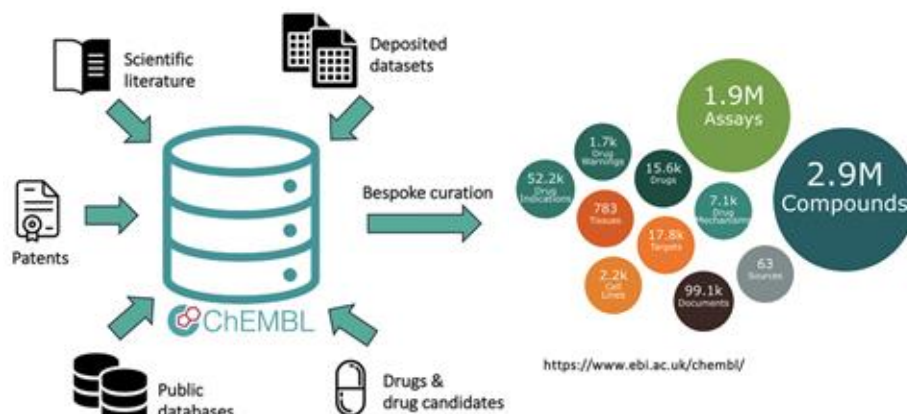
*Convert IUPAC
names to
structure*

Drug Discovery is Resource Intensive

Estimated cost >\$1 billion



ChEMBL Offers Drug Discovery Data



Also includes data on ADME, toxicity, for neglected tropical diseases



- Large openly accessible bioactivity resource
- Literature-derived (& donated) experimental measurements (+ assay metadata)
- Diverse data types; broad coverage in chemical and target space
- Expert manual curation & well annotated
- Cross-target biological context (50 years worth of data)
- Widely used for ML/benchmarking

Drug and Clinical Candidate Drug Data in ChEMBL

Journal of
**Medicinal
Chemistry**

pubs.acs.org/jmc

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Perspective

Drug and Clinical Candidate Drug Data in ChEMBL

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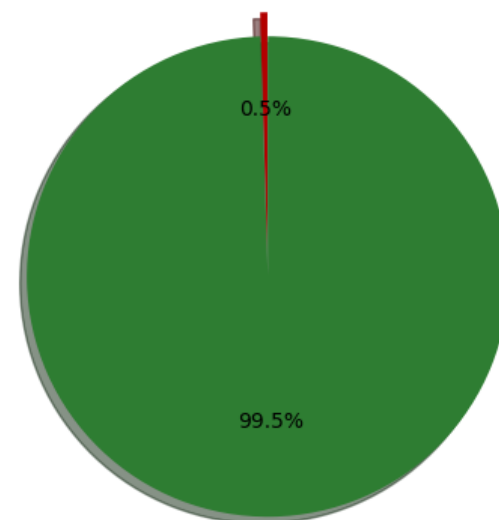
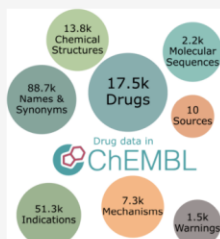
Read Online

ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: ChEMBL is a large-scale, open-access, FAIR database of bioactive molecules with drug-like properties. ChEMBL 35 contains 17,500 approved drugs, and drugs that are progressing through the clinical development pipeline. Drug curation has formed an integral part of the core offering of the ChEMBL database since its inception. The paper is a reference guide to present the principles of why the ChEMBL drug data has been curated in a particular manner so that data users can better understand the nature of the data. The drug data include information on: names, synonyms and trade names, chemical structure or biological sequence, data sources, indications, mechanisms, warnings and drug properties such as maximum phase of development, type of molecule, prodrug status and first approval. The integrated nature of the drug data within the context of a bioactivity resource enables the wide use of the data set in drug discovery, AI and machine learning.



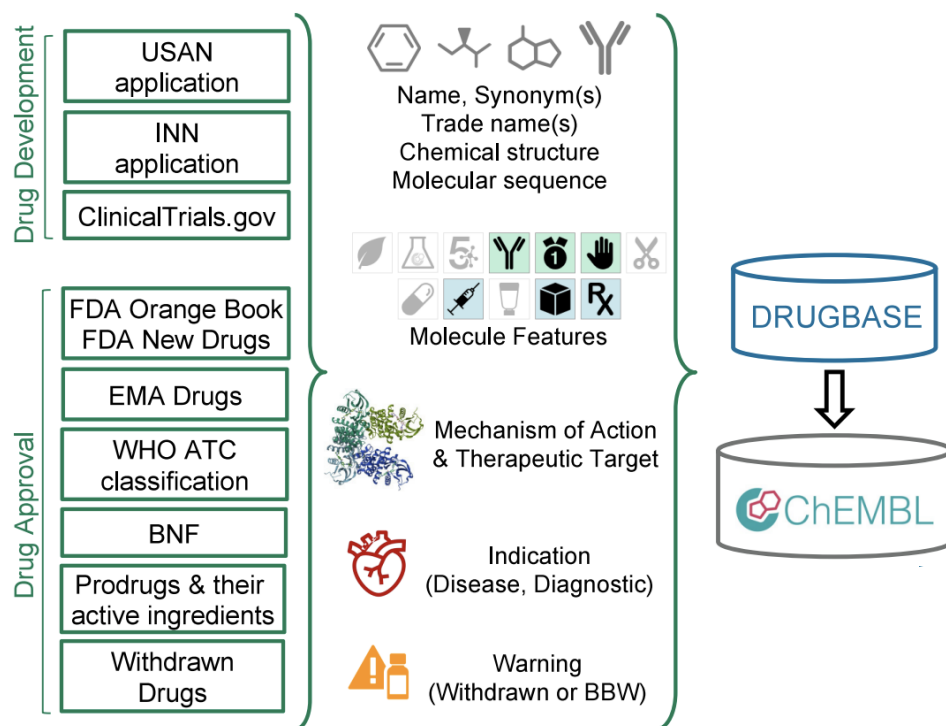
ChEMBL Literature and Deposited Compounds

ChEMBL Drug and Clinical Candidate Drug Data

Although drug data represent a small fraction of ChEMBL, they are extensively curated and form a key high-confidence dataset!

ChEMBL 35	Defining feature for inclusion in ChEMBL	Typical features in ChEMBL	Occasional or absent features in ChEMBL
Research Compound (~2.4 million)	Must have bioactivity data.	Usually measured in one or multiple assay(s) against one or multiple target(s). Usually comes from scientific literature, a data set that has been directly deposited into ChEMBL ²⁰ or patent information.	Usually does not have a preferred name (<i>pref_name</i>). Sometimes has indication, safety warning or mechanism information (if also a drug).
Clinical Candidate Drug (~14 k)	Must come from a source of clinical candidate information (e.g., USAN, INN, Clinical-Trials.gov).	Has a <i>pref_name</i> , usually a recognizable drug name or a research code. Around 40% of clinical candidate drugs have bioactivity data. (i.e., it is also a research compound). May be a prodrug.	Does not have safety warning data. May have indication and mechanism information.
Approved Drug (~4 k)	Must come from a source of approved drug information (e.g., FDA, WHO ATC, EMA).	Has a <i>pref_name</i> , usually a recognizable drug name. Around 70% of approved drugs have bioactivity data (i.e., it is also a research and/or clinical candidate compound). Usually has indication and mechanism information.	May have safety warning information (withdrawn status or black box warning). May have metabolism data (for selected drugs). May be a prodrug. May have orphan status for rare diseases.

ChEMBL Contains Multiple Sources, Molecule Types, and Other Flags / The Compound Report Card



(a) Overview of drug curation

Compound Registration & Drug Data Integration

- Unique identifier: **molregno**
 - Stored in **molecule_dictionary** (one entry per compound)
 - **compound_records** links each molregno to multiple data sources
 - BUT one compound, multiple origins (FDA, EMA, USAN, etc.)
- Structure validation & standardisation (V2000 molfile):
 - Approach varies depending on data source
- Structures may come from:
 - **USAN / INN** → **always provided** (manually drawn)
 - **FDA / EMA** → **extracted from drug labels**
 - **ATC / ClinicalTrials** → **no structure** → **name-based mapping**
 - ChEMBL Structure Pipeline (RDKit-based)
- **InChIKey matching**
 - → existing molregno or new registration
- **Biotherapeutics** (automated name mapping or manual curation)

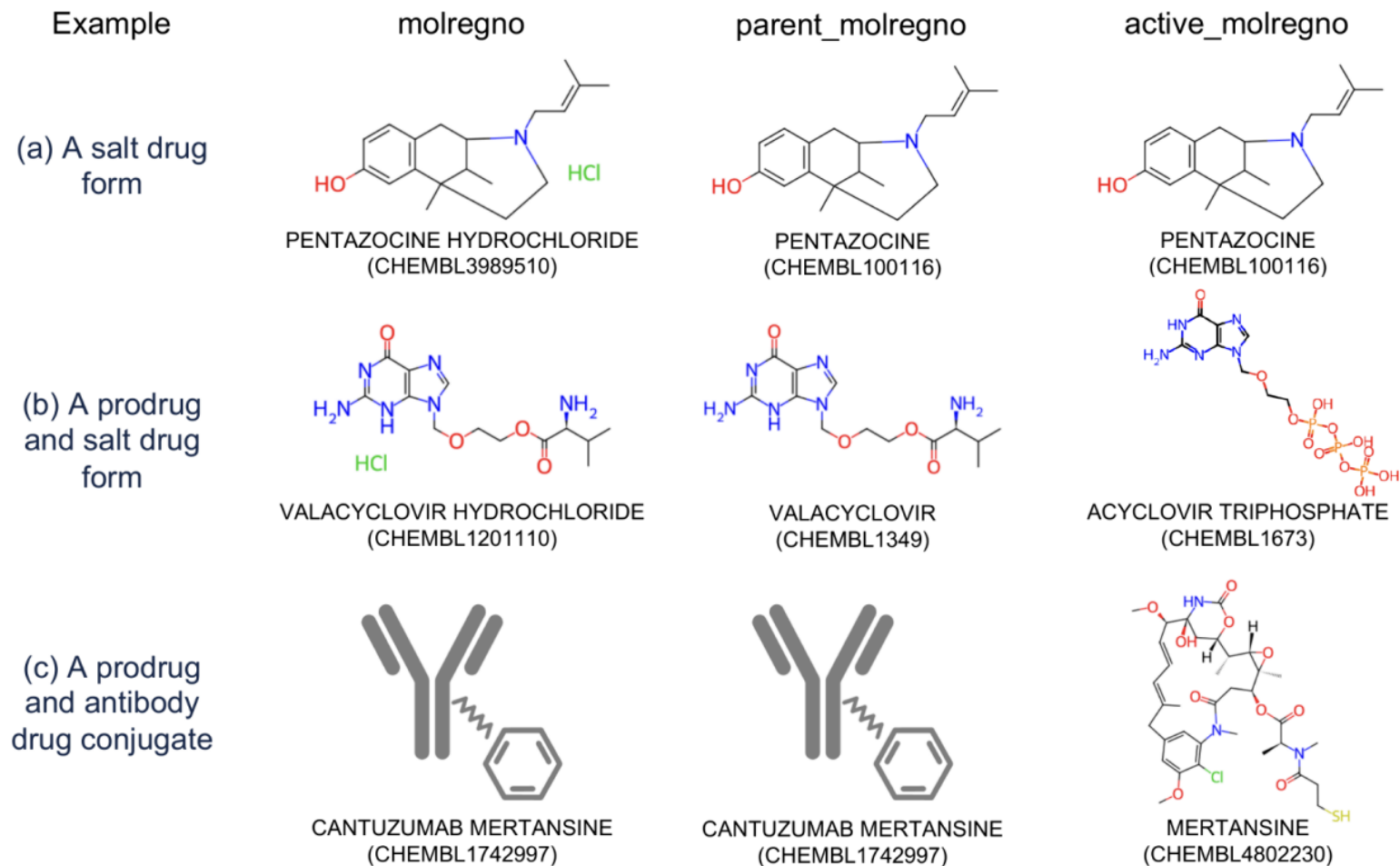
- **Preferred name (pref_name) rules:**
 - **Approved drugs** → **regulator name (FDA/EMA)**
 - **If not approved:**
 - USAN (priority)
 - INN
 - FDA GSRS
- **Challenges addressed**
 - Salt vs parent ambiguity (e.g. *abacavir sulfate* vs *abacavir*)
 - Multiple synonyms for same compound
 - Legacy issue: same name assigned to multiple forms
 - **Ongoing effort**
 - Cleanup of duplicate synonyms and names
 - Assign **distinct names to salt vs parent forms**

But are All Structures Shown in ChEMBL?

Chemical structure not displayed	Reason	Example
Mixture of compounds	If the proportions of distinct compounds within a mixture are not accurately known, then the chemical structure is not drawn e.g., a 1:1 mixture would be drawn but a compound described as “contains a mix of alpha and beta isomers...” would not be drawn.	Nystatin (CHEMBL5465559) is composed of three distinct compounds in non-specified proportions (components: A1, A2, A3 ⁵⁸) and therefore a ‘small molecule’ icon is displayed in the web interface instead of the chemical structure, 
Polymers	Polymers are not always displayed because a fully defined chemical structure is required.	Colesevelam hydrochloride (CHEMBL1201473) is a polymer ⁵⁹ with an incomplete definition of its repeating sub-units and therefore a ‘polymer’ icon is displayed in the web interface instead of a chemical structure. 
Organometallic complexes	Standard V2000 molfiles do not support co-ordinate (dative) bonds, and therefore the chemical structure information for these compounds is removed from the <i>compound_structure</i> table, and the ChEMBL interface shows an icon for a ‘metal-containing structure’. However, the molecular formula remains within the downloadable database and is shown on the ChEMBL web interface. Similarly, any associated bioactivity or other curated data remains unchanged. This approach applies to any compound that contains an element listed in (exclude_flag.py) ²¹ . This approach will be reviewed in the future as part of the plan to move towards V3000 molfiles.	Cisplatin (CHEMBL11359) is a co-ordinate complex containing platinum ⁶⁰ . Therefore a ‘metal-containing’ icon is displayed in the web interface instead of the chemical structure. 
Biotherapeutic compounds	If the chemical structure is not accurately known (e.g., antibody, protein etc.) then the chemical structure is not displayed. See Molecule Type for further information. However, biotherapeutic compounds with a small number of amino acid groups (usually < 15) may show a chemical structure in ChEMBL.	Givosiran sodium (CHEMBL4594265) is an oligonucleotide ⁶¹ and therefore an ‘oligonucleotide’ icon is displayed in the web interface. 

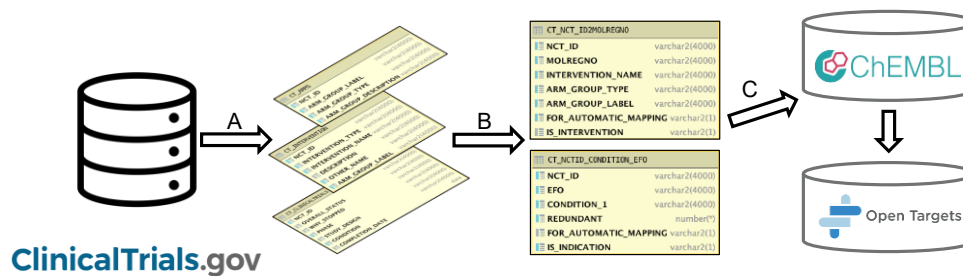
Molecule Hierarchy

ChEMBL uses the concept of compound 'families' so that each molecule, ie salts vs parent, isotopes, etc. and all the compounds in the family can be considered to have the same bioactivity.



The Clinical Trials Pipeline: An Overview

- The Clinical Trials Pipeline is the primary source of clinical trials data extracted from <https://clinicaltrials.gov/>.
- The pipeline is semi-automated and maps interventions to ChEMBL compounds and conditions to EFO IDs.



Completed ⓘ

Study Evaluating Orally Administered AMG 900 in Adult Subjects With Acute Myeloid Leukemia

ClinicalTrials.gov ID ⓘ NCT01380756

Sponsor ⓘ Amgen

Information provided by ⓘ Amgen (Responsible Party)

Last Update Posted ⓘ 2017-11-06

The Clinical Trials Pipeline: Curation Example

Study Overview

Brief Summary

This study of AMG 900 will be conducted in two parts: dose escalation and dose expansion. The dose escalation part of the study is aimed at evaluating the safety, tolerability and PK (pharmacokinetics) of oral AMG 900 in subjects with acute myeloid leukemia. Up to 93 subjects may be enrolled in dose escalation. The dose expansion part of the study will consist of 20 subjects with acute myeloid leukemia. The dose of AMG 900 will be dependent upon data from the dose escalation part of the study.

Official Title

A Phase 1 Study Evaluating the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of Orally Administered AMG 900 in Adult Subjects With Acute Myeloid Leukemia

Conditions

Cancer Hematologic Malignancies Leukemia Myeloid Leukemia

Intervention / Treatment

- Drug: Arm 1- Dose Escalation
- Drug: Arm 2- Dose Expansion

Other Study ID Numbers

- 20101351

Study Start (Actual)

2011-10-04

Primary Completion (Actual)

2014-09-04

Study Completion (Actual)

2014-09-04

Enrollment (Actual)

35

Study Type

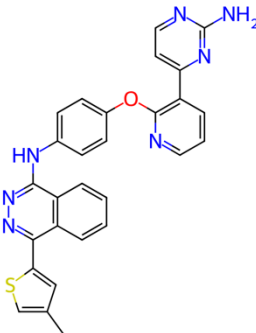
Interventional

Phase

Phase 1

AMG-900

Name and Classification



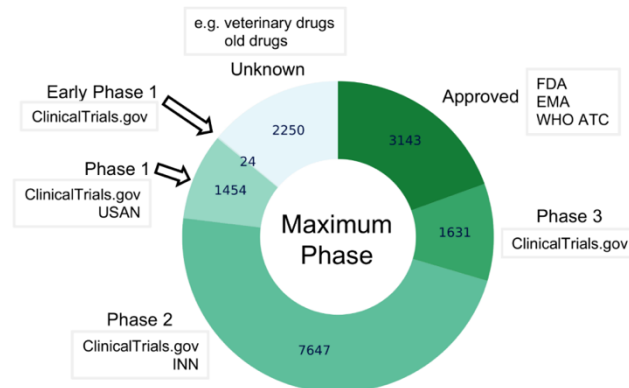
	NCT_ID	EFO	CONDITION_1	REDUNDANT	FOR_AUTOMATIC_MAPPING	IS_INDICATION
1	NCT01380756	EFO:0000565	Leukemia	1 (null)	1	
2	NCT01380756	MONDO:0004992	Hematologic Malignancies	1 (null)	1	
3	NCT01380756	MONDO:0004992	Cancer	1 (null)	1	
4	NCT01380756	MONDO:0004643	Myeloid Leukemia	(null) (null)	1	

Max Phase in ChEMBL

max_phase	Definition	Criteria / Data Source	Key Notes
4.0	Approved drug	Regulatory approval (FDA, EMA human, MHRA, PMDA)	Includes withdrawn drugs (must have been approved)
		ATC (if no other regulatory source in family) 	ATC treated as proxy for approved drugs
		Parent inherits from approved salt/form	Aggregation across compound family
3.0	Late-stage clinical candidate	Phase 3 clinical trials	From Clinical Trials Pipeline
2.0	Mid-stage clinical candidate	Phase 2 or Phase 2/3 trials	From Clinical Trials Pipeline
		INN applications	Typically submitted around Phase 2
1.0	Early clinical candidate	Phase 1 or Phase 1/2 trials	From Clinical Trials Pipeline
		USAN applications	Typically applied during Phase 1–2
0.5	Very early clinical candidate	Early Phase 1 trials	From Clinical Trials Pipeline
-1	Unknown	No assignable development phase	e.g. veterinary drugs, legacy compounds
NULL	Not a drug	Research compounds only	Bioactivity data but no clinical status



Some cmpds are very old, 1960s or earlier and approval evidence is hard to find because of the cmpds are not in use anymore. Cases like this will be corrected in ChEMBL 37 and this is work in progress.



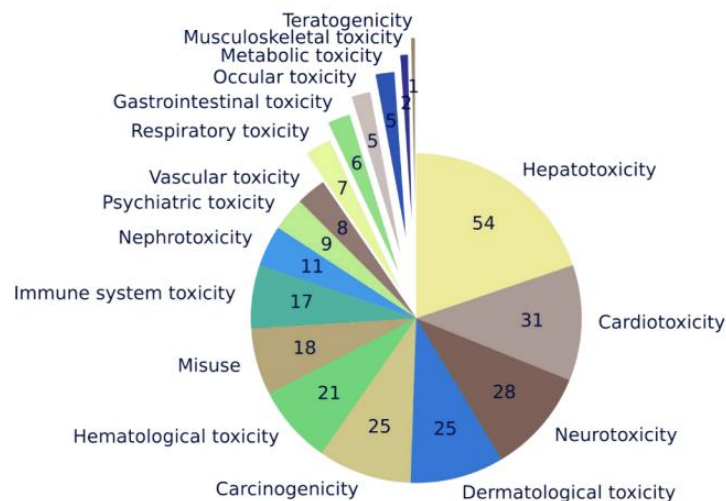
Withdrawn Drugs

- Fully manually checked and updated (withdrawn flag)
- In addition, the black box warning flag is annotated
- The drug is withdrawn at all doses
- The drug is withdrawn for all populations (and not just infants).
- The drugs is withdrawn for all indications.
- Any drug withdrawn for a lack of evidence of efficacy is not included.
- Any drug withdrawn for drug-drug interactions is included if it is a safety-related withdrawal.

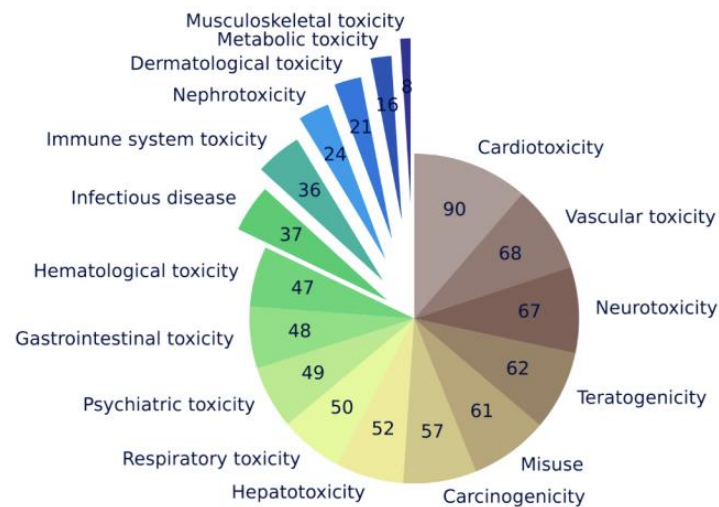
CELECOXIB (not withdrawn for all indications)

DIETHYSTILBESTROL (still used for prostate cancer)

nomifensine maleate: ChEMBL1256842



(a) Withdrawn drugs



(b) Black box warnings for approved drugs

Mechanism of Action (MOA)

- Links a drug or clinical candidate to its **therapeutic target(s)** responsible for disease effect
- Stored in **drug_mechanism** and **mechanism_refs** tables (curated with supporting evidence such as FDA/EMA labels)
- Manually curated
- Also includes **non-molecular categories** when mechanism is unclear: *diagnostic, antioxidant, laxative, unknown*
- One drug can have **multiple targets**, each with its own MOA
 - **Different targets for different indications** (e.g., imatinib: BCR-ABL vs KIT)

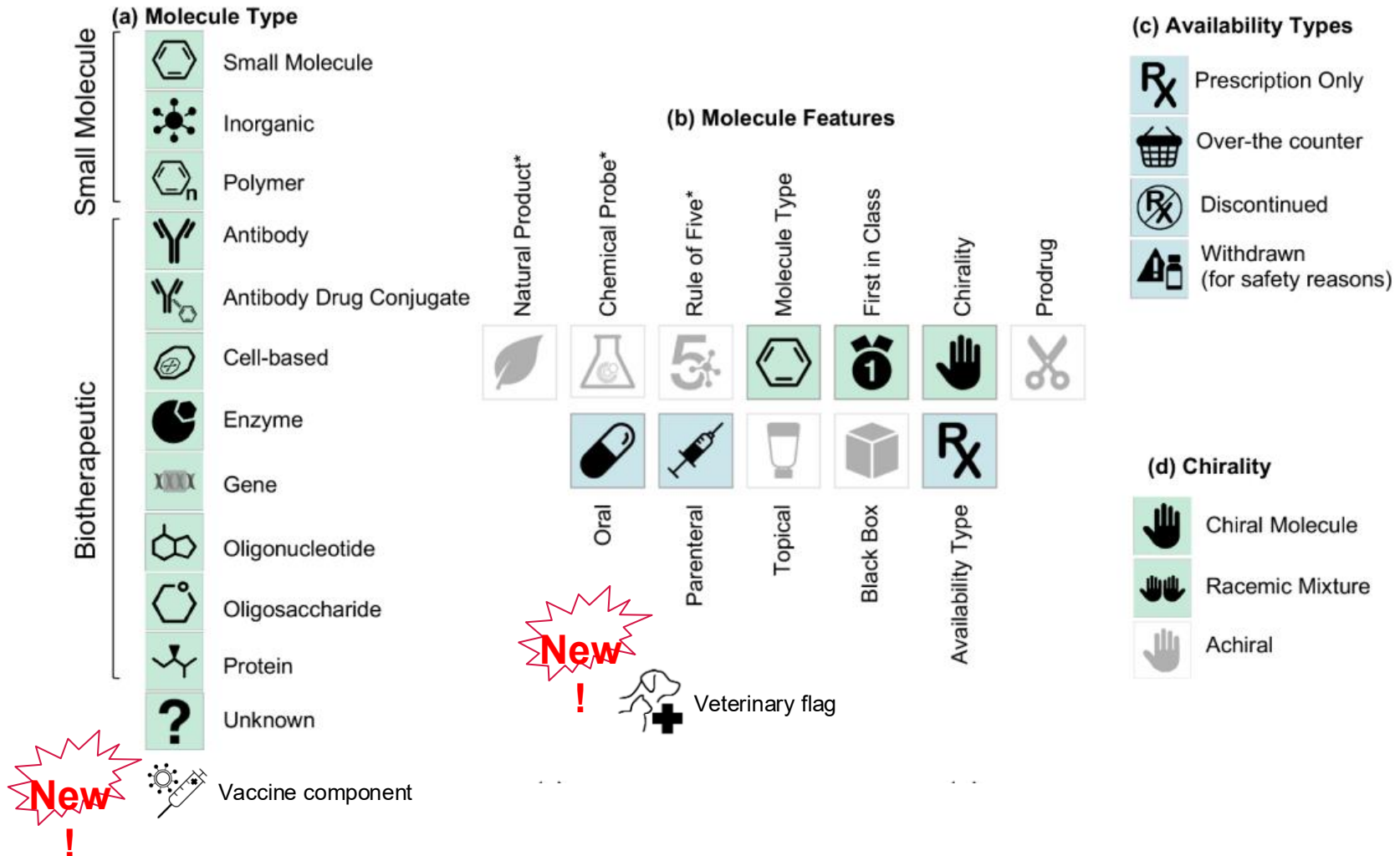
Drug Mechanisms

Items per page: 5 [BROWSE ALL](#) Showing 1-4 out of 4 records

#	Mechanism of Action	Action Type	Target ChEMBL ID	Target Name	Target Type	Target Organism	Variant	Binding Site Name	References		
1.	Tyrosine-protein kinase ABL inhibitor	INHIBITOR	CHEMBL1862	Tyrosine-protein kinase ABL1	SINGLE PROTEIN	Homo sapiens	---	---	From	DailyMed	Wikipedia
									CHEMBL1642 (Alt. Form)	[1]	[1]
2.	Stem cell growth factor receptor inhibitor	INHIBITOR	CHEMBL1936	Mast/stem cell growth factor receptor Kit	SINGLE PROTEIN	Homo sapiens	---	---	From	DailyMed	Wikipedia
									CHEMBL1642 (Alt. Form)	[1]	[1]
3.	Platelet-derived growth factor receptor beta inhibitor	INHIBITOR	CHEMBL1913	Platelet-derived growth factor receptor beta	SINGLE PROTEIN	Homo sapiens	---	---	From	DailyMed	Wikipedia
									CHEMBL1642 (Alt. Form)	[1]	[1]
4.	Bcr/Abl fusion protein inhibitor	INHIBITOR	CHEMBL2096618	Bcr/Abl fusion protein	CHIMERIC PROTEIN	Homo sapiens	---	---	From	FDA	
									CHEMBL1642 (Alt. Form)	[1]	

Showing 1-4 out of 4 records

Drug Property Information (I)



Drug Property Information (II)

Field	Definition	Encoding / Values	Key rules & curation logic	Scope / Notes
dosed_ingredient	Indicates whether this specific chemical form is administered to patients	1 = yes, 0 = no (default)	Only applied to approved human drugs. Refers to actual administered form (often salt form). Not propagated to parent molecule level.	Human approved drugs only
molecule_type	Broad classification of drug modality	small molecule, inorganic small molecule, polymeric small molecule, antibody, ADC, enzyme, protein, gene, oligonucleotide, oligosaccharide, cell-based, and vaccine component	Might be based on MW thresholds. Determined Using strycture (if available), name patterns (e.g. -mab, -ase), substructure seaches (SMILES), GSRS checks, and manual curation. Final assignment is single category per compound.	Drugs + clinical candidates
first_in_class	First approved drug for a mechanism of action at a given target	1 = yes, 0 = no, -1 = research compound	Manual curation using FDA/EMA approvals. Based on target mechanism novelty, not route, indication, or modality. New formulations or different molecule types acting on same target do not qualify if not first.	Approved drugs only
chirality	Stereochemical classification of molecule	2 = achiral, 1 = single defined stereoisomer, 0 = mixture, -1 = unknown	Determined by stereocenter presence, symmetry, and configuration certainty. Uses structure inspection, naming conventions (R/S, DL, ±), InChIKey patterns, and manual review. Racemates and mixed stereoisomers grouped under 0.	Drugs + clinical candidates
prodrug	Indicates if compound is metabolised into an active drug	1 = yes, 0 = no, -1 = research compound	Includes ester prodrugs, ProTides, and activation-dependent compounds. Only one primary active metabolite stored (active_molregno). All salt forms of a prodrug family are also flagged.	Drug metabolism annotation
route (oral/parenteral/topical)	Route(s) of administration	1 = present, 0 = absent	Based mainly on FDA Orange Book + manual curation. Multiple routes can apply. Parenteral includes IV/IM/SC/intrathecal.	Approved drugs only

Challenges in Drug Data Curation

Disparate data structures

Each source needs tailored ingestion pipelines; often requires semi-automated extraction + manual curation to ensure accuracy

Inconsistent source information

Requires cross-checking and remapping to correct drug form using multiple evidence types (e.g., trade names, labels); manual correction essential

Ambiguous compound identification

Name-based mapping (synonyms, trade names, codes) can be unreliable; same name may refer to multiple structures → ongoing deduplication and validation needed

Long-term sustainability

Need to balance scalability with human-in-the-loop curation; maintain high quality while improving efficiency

Scale of usage pressure

Increases demand for robust pipelines, automation, and infrastructure resilience without compromising quality

ChEMBL UGM

Public event/Ext Seminar event | Conference

2nd ChEMBL User Group Meeting



Scan for
more
details!



Date: 10 - 11 Jun 2026

Time: 9:00 am - 4:30 pm

Timezone: Europe/London

Location: United Kingdom, Virtual

Venue: EMBL-EBI Kendrew Theatre, and
Garden room

Deadline(s):

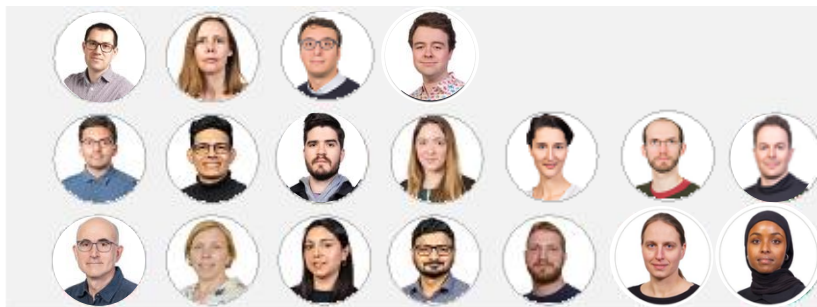
Registration (On-site): 18 May 2026

Acknowledgements



Chemical Biology Services Team

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Thank you!

